Cross-Domain Transfer of Structural Selection

A Frozen-Weight Two-Domain Pilot Study

Christof Krieg
MAAT Project – Independent Research
https://maat-research.com

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Abstract

We test whether a structural-selection architecture calibrated in one domain can transfer to a second domain without target-domain retuning. The experiment freezes a common five-support architecture $(H, B, S, V, R_{\mathrm{rob}})$ on a source domain and evaluates the resulting score on a target domain using a predeclared quality convention. Two previously generated benchmarks are used: random 3-SAT hardness data from the MAAT frustration-field benchmark and non-commuting Lindblad pointer-state data from the quantum pointer-selection benchmark.

The transfer is intentionally conservative. Sector weights are fitted only on the source domain by non-negative regression against source-domain structural cost, then frozen. The target-domain score direction, metrics, and shuffled-defect null models are declared before evaluation.

The result is asymmetric and should be interpreted as a two-domain benchmark, not as independent evidence for universal structural transfer. SAT-trained weights transfer strongly to the quantum pointer benchmark ($\rho_{\mathrm{S}} = 0.6034$), exceeding equal-weight and scalar MAAT baselines ($\rho_{\mathrm{S}} = 0.3771$) and lying far above shuffled-defect nulls. The reverse transfer from quantum to SAT does not produce positive predictive correlation ($\rho_{\mathrm{S}} = -0.0408$), although it improves over the global scalar baselines ($\rho_{\mathrm{S}} = -0.2683$).

Thus the present paper does not claim universal transfer. It demonstrates a reproducible internal transfer protocol in which connectivity-dominated SAT structure transfers strongly to quantum pointer robustness, while the reverse direction fails. The dominant transferred signal is the connectivity support V, rather than a balanced multi-sector architecture.

1 Purpose

Previous papers in this series tested structural-selection diagnostics within individual domains. This paper asks a stricter question:

> *Does a defect architecture learned in one domain retain information in a different domain without retuning?*

This is a harder test than in-domain prediction. If a framework only works after reweighting every new domain, it may be a useful feature-engineering language but not a transferable structural prior. The aim here is not to prove universality, but to convert the universality claim into a concrete falsifiable protocol. The protocol is intentionally adversarial: if transfer collapses under frozen weights, the structural-selection hypothesis loses support. The scope is deliberately narrow: two existing internal benchmark domains are used, and the result should be read as a pilot transfer audit rather than a community-level external validation.

2 Frozen Architecture

The common architecture uses the support variables

$$H, \quad B, \quad S, \quad V, \quad R_{\text{rob}}. \tag{1}$$

Here R_{rob} is the emergent robustness support used in the v1.2.1 closure, not an independent primitive sector weight. Operationally, the supports used here have the following meanings:

Support	Operational role in this transfer test
H	consistency support: low equation, rule, or target-alignment defect
B	balance support: low imbalance among competing channels or constraints
S	activity support: controlled nontrivial dynamical or structural richness
V	connectivity support: coherent coupling, graph/interference structure, or channel alignment
R_{rob}	emergent robustness support: closure-level stability/adherence score derived from the supports

These definitions are intentionally compressed. The detailed domain-specific measurements are inherited from Papers 50b and 51; the present paper tests whether the resulting shared support layer carries transferable signal.

For transfer we convert supports into support defects:

$$D_a(X) = -\log(\epsilon + \Gamma_a(X)), \qquad a \in \{H, B, S, V, R_{\text{rob}}\}. \tag{2}$$

Here $\Gamma_a \in (0, 1]$ is a support score, so D_a is a non-negative support-defect up to the small numerical offset ϵ. For a source domain A, non-negative weights are fitted only on the source:

$$\lambda^{(A)} = \arg\min_{\lambda_a \geq 0} \left\| D^{(A)}\lambda - \left(1 - q^{(A)}\right) \right\|_2^2, \qquad \sum_a \lambda_a^{(A)} = 1, \tag{3}$$

where $q^{(A)}$ is the predeclared source-domain quality. The target-domain score is then

$$\boxed{S_{A\to B}(X) = -\sum_a \lambda_a^{(A)} D_a^{(B)}(X).} \tag{4}$$

The sign convention is fixed before evaluation: higher $S_{A\to B}$ means higher structural quality. No sign flip or target-domain retuning is allowed.

3 Domains and Quality Conventions

3.1 SAT Domain

The SAT data are taken from the Paper 50b frustration-field experiment. They contain 550 random 3-SAT instances with local graph/frustration features, MAAT supports, scalar MAAT scores, and DPLL node counts. This is a deliberately limited hardness proxy. DPLL was chosen because it is transparent, deterministic, and easy to reproduce as a controlled first target, but it is not a state-of-the-art SAT hardness measure in 2026. Modern CDCL solvers with learned clauses, restarts, and VSIDS-like heuristics can induce different hardness landscapes.

The predeclared quality is

$$q_{\mathrm{SAT}} = 1 - \mathrm{minmax}\left(\log N_{\mathrm{DPLL}}\right). \tag{5}$$

Thus easier instances are treated as higher structural-quality examples. This convention does not claim that ease is equivalent to physical stability; it only defines the direction of the transfer score.

3.2 Quantum Pointer Domain

The quantum data are taken from the Paper 51 non-commuting Lindblad pointer benchmark. They contain 3500 synthetic open-system samples with MAAT supports and a pointer-robustness / fidelity target.

The predeclared quality is

$$q_{\mathrm{Q}} = \mathrm{minmax}\left(y_{\mathrm{pointer}}\right). \tag{6}$$

Higher pointer robustness is therefore treated as higher structural quality.

3.3 Internal-Artifact Caveat

Both domains are synthetic or derived artifacts generated by earlier papers in the same series. The test therefore probes internal transfer consistency under frozen weights, not generalisation to an independent external dataset.

4 Null Models

Two null families are used.

Shuffled-defect null. Each target-domain defect column is shuffled independently. This preserves each sector's marginal distribution but destroys sample-level sector coherence.

Lambda-permutation null. The frozen source weights are randomly permuted across sectors. This preserves the same coefficient values but destroys the source-learned sector assignment.

The primary transfer metrics are Spearman rank correlation and mutual information between the frozen score and target quality. The shuffled-defect null is the main falsification check because it tests whether sample-level defect coherence matters.

5 Results

5.1 Frozen Source Weights

The learned source-only weights are:

Table 1: Frozen source-only sector weights.

Source	λ_H	λ_B	λ_S	λ_V	$\lambda_{R_{\mathrm{rob}}}$
SAT	0.0000	0.0000	0.0000	1.0000	0.0000
Quantum	0.4839	0.0000	0.2978	0.2183	0.0000

The SAT source architecture collapses to a connectivity-dominated mode. This is not imposed manually. It is the result of the source-only non-negative fit. The quantum source architecture distributes weight over consistency, activity, and connectivity.

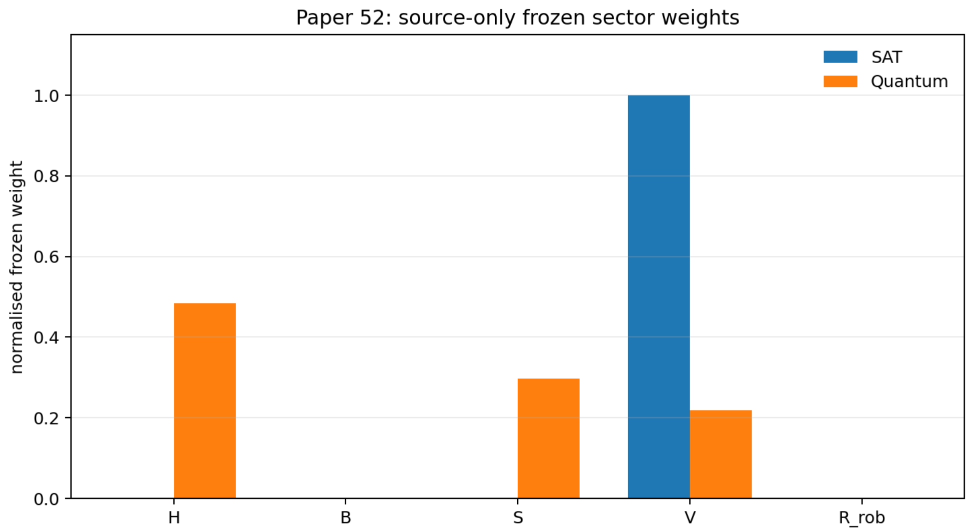

Figure 1: Frozen sector weights learned on the source domains only.

5.2 Transfer Performance

Table 2: Cross-domain transfer results. Higher Spearman correlation means better alignment with the predeclared target-domain quality.

Transfer	Frozen MAAT	Equal/scalar MAAT	$\Delta\rho_S$
SAT → Quantum	0.6034	0.3771	+0.2263
Quantum → SAT	-0.0408	-0.2683	+0.2276

The transfer is asymmetric. SAT-trained connectivity transfers strongly to the quantum pointer benchmark. The reverse transfer does not yield positive SAT-hardness prediction, although it is less negative than the equal-weight and scalar MAAT baselines. Because the SAT-trained fit collapses to $\lambda_V = 1$, the positive direction should be interpreted carefully. It may indicate a transferable structural prior, but it may also reflect the fact that both chosen benchmarks contain graph-like or connectivity-like structure. This ambiguity is a limitation of the present two-domain design.

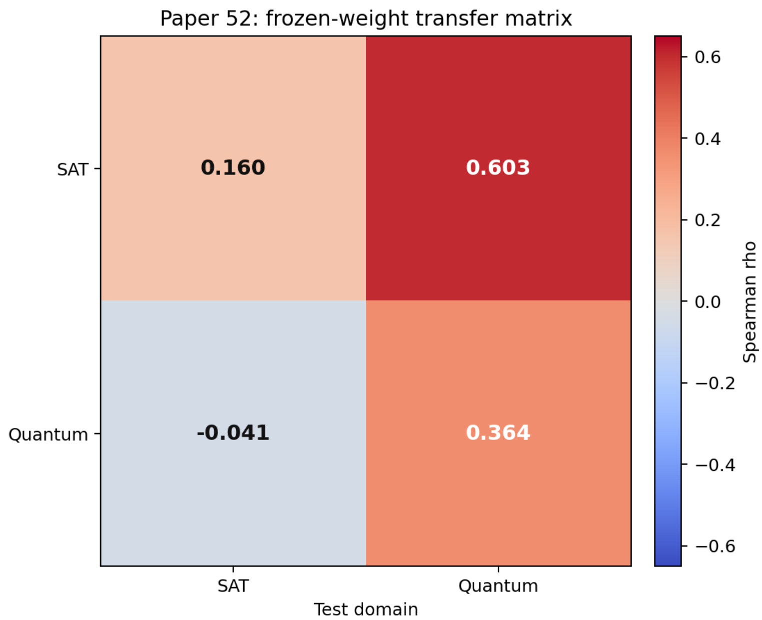

Figure 2: Frozen-weight transfer matrix. Rows indicate the source domain used to learn frozen sector weights; columns indicate the test domain. Diagonal entries are included only for orientation and are not independent held-out validation.

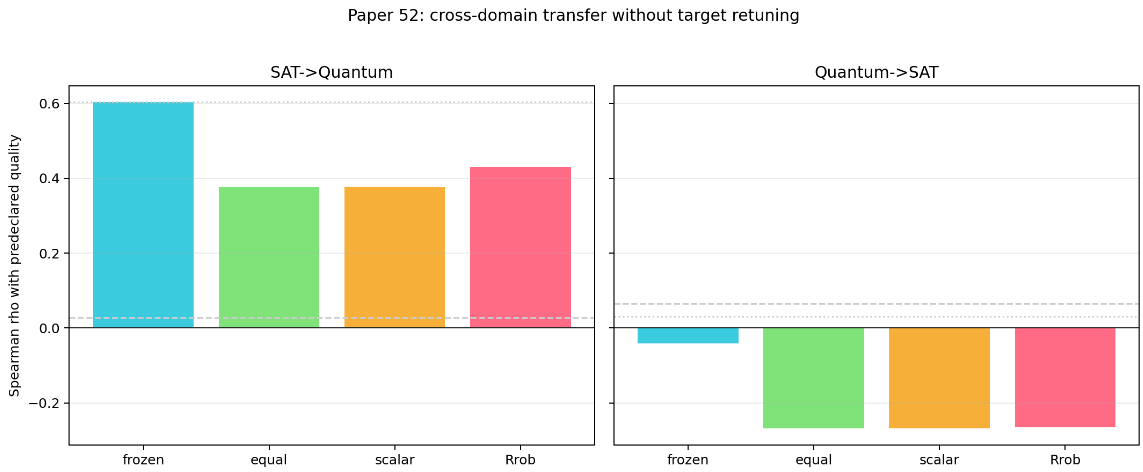

Figure 3: Transfer Spearman correlations for frozen MAAT, equal-weight MAAT, scalar MAAT, and R_{rob}-only baselines. Grey reference lines show null quantiles.

5.3 Shuffled-Null Validation

For SAT $\to$ Quantum, the frozen score is far above the shuffled-defect null:

$$\rho_{\text{real}} = 0.6034, \qquad \rho_{\text{shuffle},95\%} = 0.0272, \qquad p_{\text{shuffle}} \simeq 0.002. \tag{7}$$

The mutual-information result is similarly above null:

$$I_{\text{real}} = 0.3371, \qquad I_{\text{shuffle},95\%} = 0.0161, \qquad p_{\text{shuffle}} \simeq 0.002. \tag{8}$$

For Quantum $\to$ SAT, the frozen score does not beat the shuffled Spearman null:

$$\rho_{\text{real}} = -0.0408, \qquad \rho_{\text{shuffle},95\%} = 0.0644. \tag{9}$$

However, its mutual information remains above the shuffled-null upper tail:

$$I_{\mathrm{real}} = 0.0463, \qquad I_{\mathrm{shuffle},95\%} = 0.0402. \tag{10}$$

This indicates weak residual information but not a successful positive rank-transfer result.

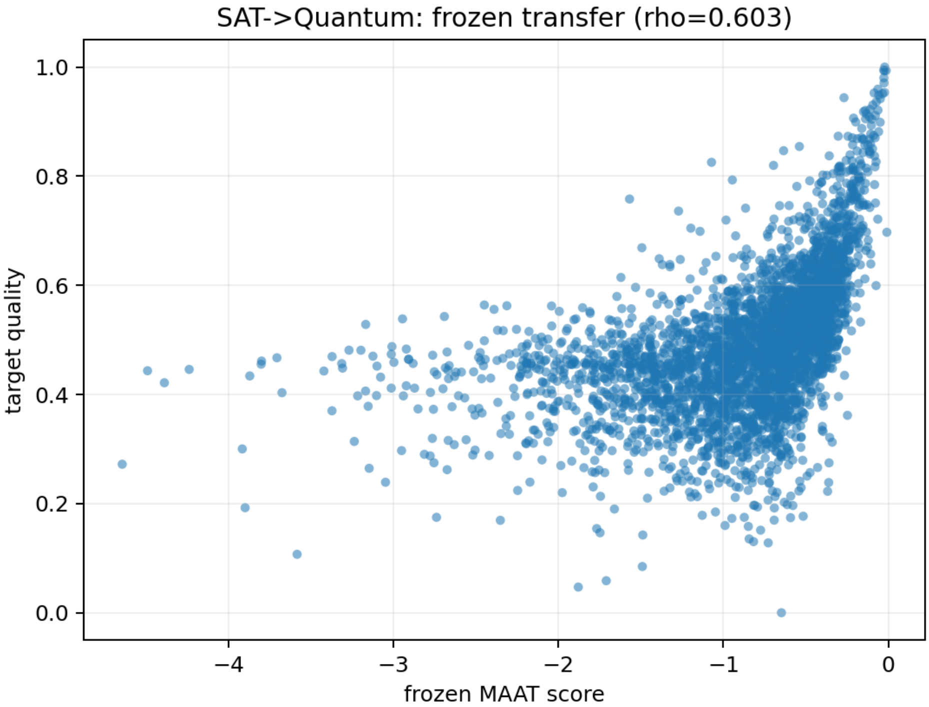

Figure 4: SAT → Quantum transfer. The source-frozen score aligns strongly with quantum pointer quality.

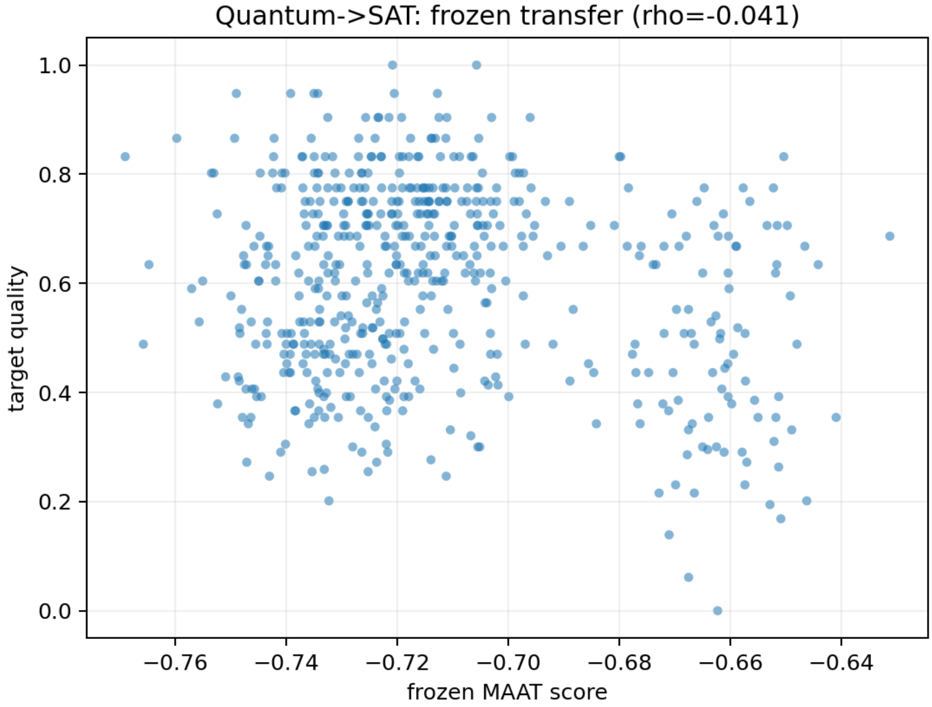

Figure 5: Quantum → SAT transfer. The source-frozen score does not produce positive rank alignment with SAT structural quality.

6 Interpretation

The result should be read as a partial transfer success and a partial falsification. It is not evidence that a single scalar MAAT score transfers universally. Indeed, the SAT direction continues to confirm the earlier lesson from Papers 50–50c: SAT hardness is sensitive to local frustration geometry and loses information under aggressive scalar compression.

> *The experiment suggests that low-frequency structural modes may transfer across domains even when high-frequency domain-specific defect geometry does not.*

The positive SAT → Quantum result is nevertheless important. The source-only fit selects the connectivity support V, and this same connectivity-dominated architecture is informative for quantum pointer robustness. The conservative interpretation is that the two internal benchmarks share a connectivity-sensitive structural mode. This suggests that low-frequency structural signals, particularly connectivity, can be partially transferable even across conceptually distant domains, at least within the current MAAT support language. Whether this should be called a universal structural prior or simply a shared graph-like feature of the two benchmarks cannot be decided from the present data alone.

The reverse failure is equally informative. Quantum pointer robustness does not provide a frozen architecture that solves SAT hardness. The asymmetric failure is scientifically valuable because it demonstrates that the protocol can fail under frozen transfer. This supports the more modest v1.4 interpretation:

> *global MAAT supports may transfer as low-frequency structural modes, but hard domains may require domain-specific defect-field resolution.*

7 Relation to the v1.4 Mode Interpretation

The present result is naturally aligned with the v1.4 defect-field view. The earlier scalar reading of MAAT could be interpreted as a search for one global transferable score. Papers 50–50c already showed that this is too coarse for SAT: local tails, clusters, and frustration hotspots are lost under scalar compression.

Paper 52 adds a transfer interpretation. The transferable object is not the full scalar F_{MAAT} and not a balanced multi-sector architecture. It is a specific low-frequency structural mode, namely the connectivity support V. This suggests a decomposition of the effective structural space into global and local components,

$$\mathcal{S}_{\mathrm{struct}} \simeq \mathcal{S}_{\mathrm{global}} \oplus \mathcal{S}_{\mathrm{local}}, \tag{11}$$

where global low-frequency modes may partially transfer across domains, while local high-frequency defect geometry remains domain-specific.

In this sense, the result is compatible with an approximate mode-separation interpretation of the MAAT support language: the sectors should not be treated as redundant summaries of the same scalar quantity, but as partly independent structural axes. Transfer is then expected to be sector- and scale-dependent, not all-or-nothing. This is an interpretation of the present benchmark, not an additional proof of universality.

8 Limitations

- Only two domains are tested.
- Both domains use synthetic or derived benchmark artifacts from previous papers, so the test is not independent external validation.
- The supports are compressed abstractions of earlier domain-specific measurements and may remain opaque without the companion papers.
- The source fitting rule is simple non-negative least squares, not a full Bayesian transfer model.
- The SAT domain uses DPLL node count, not modern CDCL conflict counts, restart statistics, or learned-clause dynamics.
- The quantum benchmark is a toy Lindblad ensemble, not a full quantum measurement theory.
- The successful direction is connectivity-dominated and may partly reflect shared graph-like structure rather than a domain-independent law.
- Future work should test whether the V-dominance persists under alternative fitting rules, regularisation schemes, and additional domains.
- The result is asymmetric and should not be presented as a universal transfer proof.

9 Reproducibility

The experiment is stored in:

```
experiments/cross_domain_transfer_paper52/
```

Run from the repository root:

```
python3 experiments/cross_domain_transfer_paper52/cross_domain_transfer_paper52.py
```

The script reads:

```
experiments/sat_frustration_fields_paper50b/outputs_paper50b/
experiments/quantum_pointer_state_selection_paper51/outputs_paper51/
```

and writes:

```
experiments/cross_domain_transfer_paper52/outputs_paper52/
```

The main output files are:

- `paper52_transfer_results.csv`
- `paper52_shuffle_nulls.csv`
- `paper52_frozen_weights.csv`
- `paper52_summary.json`

No new external dataset is introduced in this paper. The data are repository-internal synthetic/derived artifacts generated for the Paper 50b and Paper 51 benchmarks.

10 Conclusion

Paper 52 introduces a frozen cross-domain transfer protocol for structural selection. The protocol is deliberately stricter than in-domain prediction: weights are learned on a source domain, frozen, and evaluated in a target domain with predeclared scores and shuffled-defect nulls.

The first result is asymmetric and deliberately modest. SAT-trained structural connectivity transfers strongly to quantum pointer-state robustness, while quantum-trained structure does not solve SAT hardness. This is not a universal proof. It is a useful and falsifiable pilot result: some low-frequency structural modes can survive frozen transfer across two internal benchmarks, but hard combinatorial domains still require local multi-scale defect fields and independent external validation remains open.

References

[1] E. T. Jaynes, "Information Theory and Statistical Mechanics," *Physical Review* **106**, 620–630 (1957).

[2] C. Krieg, "A Maximum-Entropy Derivation of Universal Structural Selection," preprint (2026).

[3] C. Krieg, "Universality Benchmark Audit for Structural Selection," preprint (2026).

[4] C. Krieg, "SAT Structural Hardness IIb: Frustration-Field Compression and the Limits of Scalar Robustness," preprint (2026).

[5] C. Krieg, "Multi-Scale Defect-Field Structural Selection," preprint (2026).

[6] C. Krieg, "Quantum Pointer-State Selection in a Non-Commuting Lindblad Benchmark," preprint (2026).